# Paper Review: 3 Papers
# RL & QRL for Power Systems

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# Paper Review — Lin et al. (2025)
# DDSAC: Deep RL for Multi-Objective Transmission Switching

# Limitations of Current Research

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- **Multi-objective transmission switching is NP-hard — classical solvers fail at scale**
  - AC power flow nonlinearity + binary line switching = computationally intractable beyond ~200 buses
  - Existing MIP solvers cannot meet real-time operation requirements on large systems
- **Prior DRL methods (PPO, DDQN) had unstable convergence on multi-objective switching**
  - Global reward evaluation — impossible to attribute which specific line switch caused improvement
  - No method balanced all 5 objectives simultaneously: cost, voltage, congestion, loss, switching count
- **No feasibility guarantee — islanding and power flow divergence risk during training**
  - Only a large penalty −Re discourages bad topologies; no hard physical constraint enforced
  - Diverged power flows waste computation and significantly slow RL agent learning
- **Limited generalization — trained and tested on IEEE 118-bus only**
  - Full retraining required for any topology change, different grid size, or contingency scenario

# Contributions

- **Why apply DRL to transmission switching?**

– Classical solvers too slow for real-time control; DRL learns without pre-solved optimal solutions
– DRL adapts continuously to changing load, generation dispatch, and contingency conditions

- **DDSAC — Discrete Dueling Soft Actor-Critic (novel architecture)**

– Dueling Q-network: $Q(s,a) = V(s) + A(s,a) - \text{mean}(A)$ — isolates per-line switching contribution
– Value stream $V(s)$ captures state quality; Advantage $A(s,a)$ ranks individual switching decisions
– SAC entropy term: encourages exploration of diverse switching combinations, avoids local optima

- **Simultaneous 5-objective reward function**

– $r = -(w1 \cdot cost + w2 \cdot Vviol + w3 \cdot congestion + w4 \cdot loss + w5 \cdot Nswitch)$ with $w2 = 100$ (voltage top priority)
– Islanding penalty $-Re$ acts as soft safety constraint against grid disconnection

- **Results: converges ~50 steps; outperforms PPO and DDQN on all 4 metrics across 10 seeds**

# Key Metrics, Dataset & RL Details

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## Key Metrics

| Metric | Target / Result |
|---|---|
| Cumulative Reward | Highest & stable after ~50 steps |
| Generator Cost | Lowest vs. PPO and DDQN |
| Voltage Violation | Lowest (w2 = 100 in reward) |
| Active Power Loss (MW) | Lowest |
| Convergence Speed | ~50 steps (PPO/DDQN: 200+) |

## RL Configuration & Dataset

| Parameter | Value |
|---|---|
| Algorithm | DDSAC (Discrete Dueling SAC) |
| State space | PG, voltages, flows, losses, loads, status, t |
| Action space | Binary switch vector (186 lines) |
| Reward | −(0.1·cost+100·Vviol+1·cong+0.1·loss+sw) |
| γ / LR / batch | 0.99 / 1e-4 / 32 |
| Network / τ | 2×256 MLP / 0.005 |
| Test system | IEEE 118-bus via MATPOWER + PyTorch |

⚠ **Key Limitation:** No hard feasibility constraint · IEEE 118-bus only · No topology generalization · AC convergence risk

# Paper Review — Kim et al. (2026)
# DA-DNN: Unsupervised Learning with Embedded OPF
# for Optimal Transmission Switching

# Limitations of Current Research

- **MIP solvers become intractable on large grids — 48+ hours without a solution on 300-bus**
  - No feasible result returned within commercial solver time limits for IEEE 300-bus DC-OTS
  - Cannot support real-time or near-real-time power system operational decisions
- **Supervised learning requires prohibitively expensive pre-solved labels**
  - Solving MIP-OTS thousands of times to generate training data is infeasible at 300-bus scale
  - Existing SL methods (Pineda, Bugaje) cannot extend to large systems for this reason
- **DC approximation only — ignores reactive power, voltage magnitudes, and losses**
  - Real grids require AC power flow; DC model misses voltage violations and reactive limits
  - Extending to embedded AC-OPF layer is stated as future work by the authors
- **Single-period, fixed-topology — no temporal coordination, no generalization**
  - No intertemporal constraints (ramp rates, storage); retraining required for any new grid topology

# Contributions

- **Why unsupervised DNN + embedded OPF? — No labels needed; physics enforced end-to-end**
  - RL has no feasibility guarantee; SL needs expensive labels; neither scales to 300-bus
  - DA-DNN uses generation cost directly as training loss — the embedded OPF enforces physics
- **First unsupervised DNN for DC-OTS with differentiable embedded DC-OPF layer**
  - Network $\Phi$ predicts relaxed statuses $\hat{z} \in [0,1]$; DC-OPF layer solves dispatch given $\hat{z}$
  - KKT conditions + implicit differentiation → backprop through OPF → physics-aware training
- **Smart initialization + binary regularization guarantee feasibility throughout training**
  - Wlast=0, blast=9 → $\hat{z}$_init≈0.9999 (all-lines-in start) prevents early infeasibility
  - $R(\hat{z}) = \alpha\|\hat{z}\odot(1-\hat{z})\|_1$ pushes predictions toward 0/1 → 100% feasibility after binarization
- **300-bus solved in 10 ms; MIP fails after 48 hours — 1.70% cost reduction vs. DC-OPF**

# Key Metrics, Dataset & Method Details

## Performance vs. Baselines

| Metric | IEEE 73-bus | IEEE 118-bus | IEEE 300-bus |
|---|---|---|---|
| Avg. Gen. Cost ($k) | 183.01 (= DC-OPF) | 93.02 (MIP: 93.01) | 514.00 (DC-OPF: 522.94) |
| Constraint Violations | 0% | 0% | 0% |
| Inference Time (s) | 0.00 ± 0.00 | 0.00 ± 0.00 | 0.01 ± 0.00 |
| MIP Solver Time | 0.36 ± 0.45 s | 2.27 ± 1.72 s | NA (>48 hours) |
| Feasibility (binarized) | 100% | 100% | 100% |

## Method & Loss

- Loss: $L = C(\hat{p}g^*) + \alpha\|\hat{z}\odot(1-\hat{z})\|_1$ ($\alpha = 10$)
- Optimizer: AdamW | LR: 1e-4 / 1e-3 (300-bus)
- Arch: 3-layer FC-DNN + CVXPYLayer DC-OPF
- Solver: Clarabel (training) / Gurobi v13 (test)

## Dataset

- PGLib-OPF: IEEE 73 / 118 / 300-bus
- 3,000 feasible load samples per system
- Scaling: $\kappa \sim U(0.90, 1.10) \times$ base demand
- Split: 50% train / 16.7% val / 33.3% test
- Tool: PyTorch + CVXPYLayer + Gurobi v13

# Paper Review — Lin et al. (2025)
# QE-SAC: Quantum Reinforcement Learning for Volt-VAR Control in Distribution Systems

# Limitations of Current Research

- **Classical DRL requires millions of parameters — high computational cost and low interpretability**
  - SAC baseline: 899K–3.2M parameters for 13–123-bus systems → overfitting + interpretability issues
  - Classical DRL was not designed to benefit from quantum hardware acceleration
- **Existing QRL methods cannot handle high-dimensional VVC state with limited qubits**
  - Raw grid state is too large for direct quantum encoding on NISQ-era devices
  - No prior QRL work addressed multi-device, multi-objective VVC for distribution systems
- **QE-SAC tested only on classical simulator — real quantum advantage not yet proven**
  - PennyLane default.qubit (statevector simulation) — actual quantum hardware not used
  - Simulator overhead makes wall-clock time longer than classical SAC for 13/34-bus
- **Barren plateau risk as qubit count or circuit depth grows beyond current setup**
  - Only 8 qubits and 2 layers tested; gradient vanishing may worsen for systems larger than 123-bus
  - SAC-AE baseline fails to converge — limits fair classical vs. quantum comparison

# Contributions

- **Why Quantum RL for VVC? — ~1% of classical parameters with comparable performance**

– Quantum superposition + entanglement enables compact policy in high-dimensional Hilbert space
– Multi-device coordination (capacitors, regulators, batteries) needs efficient action representation

- **QE-SAC: CAE + 8-qubit 2-layer VQC inside SAC — first QRL for multi-device VVC**

– Classical autoencoder compresses s → s′ (8-dim) → angle encoding Ry(s′i)|0⟩ into 8 qubits → VQC
– Variational layers: CNOT(nearest-neighbor) entanglement + Rx(ζk) rotations; parameter-shift gradient
– Output: linear + softmax → per-device action probability for each controllable device

- **Co-adaptive autoencoder (CAE) retrained every C=500 steps to track grid dynamics**

– Keeps the quantum latent space aligned with non-stationary load/generation changes during training

- **Zero voltage violations on 13 & 123-bus; robust under depolarizing noise λ=0.1%–1%**

– 4,872 parameters (vs 899,729 SAC) on 13-bus · 42,329 (vs 3,213,427 SAC) on 123-bus

# Key Metrics, Dataset & QRL Details

## Performance Comparison (avg. 10 seeds)

| Algorithm | 13-bus Reward | 13-bus VViol | 123-bus Reward | 123-bus VViol | Params (123-bus) |
|---|---|---|---|---|---|
| QE-SAC (ours) | −5.39 | 0 | −9.53 | 0 | 42,329 |
| Classical SAC | −5.41 | 0.01 | −9.72 | 0 | 3,213,427 |
| QC-SAC (fixed PCA) | −5.91 | 0.02 | −10.78 | 0.01 | ~42K |
| SAC-AE | N/A | N/A | N/A | N/A | ~41K |

## QRL Configuration

- Algorithm: QE-SAC (CAE-VQC inside SAC)
- VQC: 8 qubits · 2 variational layers · angle encoding Ry
- Gradient: parameter-shift rule
- Reward: $r = -[\alpha \cdot fvv + \beta \cdot fcl + \gamma \cdot fpl]$
- $\gamma$=0.99 · lr=1e-4 · batch=256 · buffer=1M · $\rho$=0.005
- CAE update every C=500 gradient steps

## Dataset

- PowerGym (OpenDSS — 3-phase unbalanced AC)
- IEEE 13 / 34 / 123-bus distribution feeders
- Episode: 24 time steps (full 24-hour horizon)
- Devices: capacitors (ON/OFF), regulators (33 taps), batteries (33 charge levels)
- Framework: PennyLane default.qubit + PyTorch

# Open Source & Dataset

- **Paper 1 — DDSAC (Lin et al. 2025)**
  (a) arXiv: https://arxiv.org/abs/2507.11726
  (b) Dataset: IEEE 118-bus via MATPOWER (simulation env, not publicly released)
- **Paper 2 — DA-DNN (Kim et al. 2026)**
  (a) arXiv: https://arxiv.org/abs/2512.17516
  (b) Dataset: PGLib-OPF → https://github.com/power-grid-lib/pglib-opf
- **Paper 3 — QE-SAC (Lin et al. 2025)**
  (a) DOI: https://doi.org/10.1109/OAJPE.2025.3534946
  (b) Dataset: PowerGym → https://github.com/siemens/powergym

# Q & A

Thank you!